CUSTOMER CHURN PREDICTION

Using IBM SPSS Modeler

A

PROJECT REPORT

SUBMITTED BY

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ABSTRACT

Customer churn is a significant challenge for organizations, particularly in competitive industries such as telecommunications, banking, and subscription-based services. High customer attrition can reduce revenue, increase acquisition costs, and weaken long-term customer relationships. Predicting customer churn in advance can help organizations understand attrition patterns and develop effective retention strategies.

The main objective of this study is to explore how predictive analytics can be used to predict customer churn with the help of IBM SPSS Modeler. The study uses a synthetic dataset containing 100 customer records and 16 attributes related to customer demographics, service usage, contract details, and billing information. The target variable, Churn, classifies customers into two categories: “Yes” and “No”.

IBM SPSS Modeler is used to import and prepare the dataset, configure the relevant variables, and develop a predictive model using the CHAID (Chi-squared Automatic Interaction Detection) Decision Tree algorithm. The model output is examined to understand how decision-tree techniques can support customer churn analysis and assist marketing and customer-success professionals in identifying patterns associated with customer turnover.

The study provides practical exposure to data preparation, predictive modelling, and the interpretation of model outputs. However, the results are preliminary because the dataset is synthetic and the model requires further validation and refinement before it can be used for real-world business decisions. The study highlights the potential of predictive analytics as a decision-support tool for Customer Relationship Management.

Keywords: *Customer Churn, Predictive Analytics, IBM SPSS Modeler, CHAID Decision Tree, Customer Relationship Management, Classification Model.*

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1. INTRODUCTION

Customer churn is one of the major challenges faced by organizations across different industries, especially in highly competitive sectors such as telecommunications, banking, insurance, and digital subscription services. It refers to the situation in which customers discontinue their relationship with a company by cancelling services, switching to competitors, or stopping purchases. High customer churn can reduce recurring revenue, increase customer acquisition costs, and negatively affect brand reputation.

In today's competitive business environment, organizations need to understand the factors that influence customers' decisions to leave. Factors such as tenure, monthly charges, contract type, payment method, internet service quality, technical support availability, and overall service satisfaction may help in understanding customer churn patterns.

Predictive analytics is a data-driven approach that uses historical data, statistical techniques, and machine learning algorithms to identify patterns and predict possible future outcomes. In Customer Relationship Management (CRM), predictive analytics can help organizations identify customers who may be at risk of leaving and take appropriate steps to improve customer retention.

This project, titled "Customer Churn Prediction Using IBM SPSS Modeler," aims to analyze customer-related data and develop a predictive model to classify whether a customer is likely to churn. The project uses a sample customer dataset containing information about customers' demographic characteristics, service subscriptions, billing details, contract terms, and usage patterns.

IBM SPSS Modeler will be used for data preparation, exploratory analysis, predictive model development, and model evaluation. The model's performance will be assessed using suitable evaluation measures, such as accuracy, precision, recall, and a confusion matrix, wherever supported by the selected modeling approach.

The findings of this project can help explain customer churn patterns and demonstrate how predictive analytics can support Customer Relationship Management. The results may provide useful insights for developing customer retention strategies, improving service quality, and supporting data-driven marketing decisions.

Note: The dataset used in this project is a synthetic sample dataset created for academic and learning purposes. Therefore, the results demonstrate the application of predictive analytics and should not be treated as evidence about any real organization.

2. PROBLEM STATEMENT

Customer churn is a significant challenge for organizations as it can lead to loss of recurring revenue, increased marketing and acquisition costs, reduced market share, and disruption of long-term customer relationships. Identifying the factors that influence customers to leave an organization is important for developing effective customer retention strategies.

Traditional approaches to customer attrition analysis may not be sufficient to identify complex patterns in customer data or estimate which customers may be at risk of leaving. Therefore, organizations can use predictive analytics techniques to analyze customer information and identify patterns associated with churn.

The main problem addressed in this project is to analyze customer-related factors, such as tenure, monthly charges, contract type, payment method, internet service, technical support, and paperless billing, to understand their relationship with customer churn.

This project aims to use IBM SPSS Modeler to prepare and analyze a sample customer dataset and develop a predictive model that classifies customers into churn categories: "Yes" or "No." The model's performance will be evaluated using appropriate classification metrics to understand its predictive capability.

The ultimate objective is to demonstrate how predictive analytics can support Customer Relationship Management in understanding customer churn patterns and making informed decisions to improve customer retention.

3. DATA DISCUSSION

3.1 Overview of the Dataset

The dataset used in this project, "Customer Churn Prediction Using IBM SPSS Modeler," contains customer-related information for analyzing factors associated with customer churn. The dataset has been prepared for academic and learning purposes to demonstrate the application of predictive analytics in Customer Relationship Management.

The dataset contains 100 customer records and 16 attributes, including demographic information, service usage characteristics, contract details, billing information, and churn status.

3.2 Source of Data

The data used in this project is a synthetic sample dataset created for academic purposes. It does not represent actual customers of any organization. The dataset is used to demonstrate data preparation, analysis, predictive model development, and evaluation using IBM SPSS Modeler.

3.3 Description of Variables

The dataset includes the following important variables:

- CustomerID: Unique identification number assigned to each customer.
- Gender: Gender of the customer (Male/Female).
- SeniorCitizen: Indicates whether the customer is a senior citizen (0/1).
- Partner: Indicates whether the customer has a partner (Yes/No).
- Dependents: Indicates whether the customer has dependents (Yes/No).
- Tenure: Number of months the customer has been with the company.
- PhoneService: Indicates whether the customer has phone service.
- InternetService: Type of internet service (DSL, Fiber optic, No).
- OnlineSecurity: Whether the customer has online security service.
- TechSupport: Whether the customer has technical support.
- Contract: Type of contract (Month-to-month, One year, Two year).
- PaperlessBilling: Whether the customer uses paperless billing.
- PaymentMethod: Method of payment (Electronic check, Mailed check, Bank transfer, Credit card).
- MonthlyCharges: Monthly charges paid by the customer.
- TotalCharges: Total charges accumulated by the customer.
- Churn: Indicates whether the customer has left, represented by Yes or No.

3.4 Target Variable

The target variable for this predictive analytics project is Churn. It has two categories: Yes and No. The predictive model will use the other relevant customer attributes to classify the target variable.

3.5 Exploratory Data Visualization

Before building the predictive model, exploratory visualizations were created to understand the distribution of the target variable and key relationships in the data.

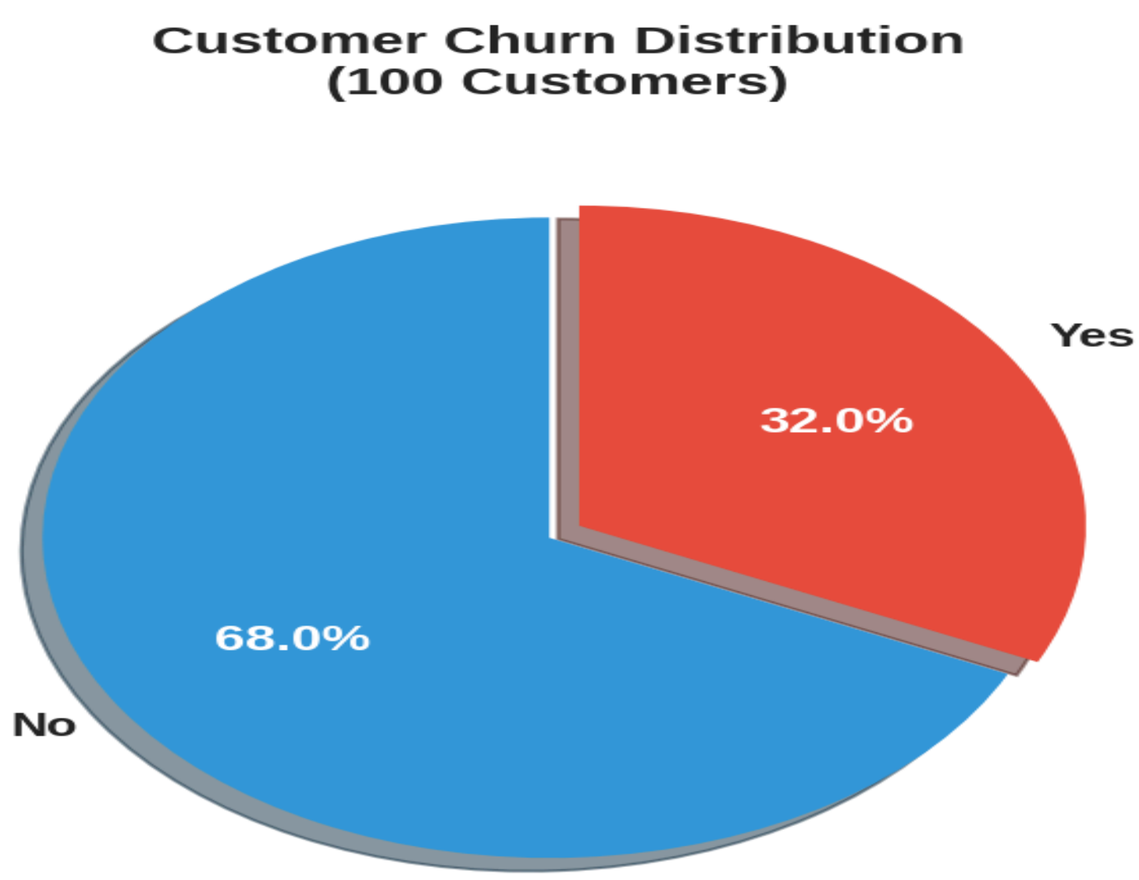


Figure 1: Distribution of Customer Churn (Yes = 32%, No = 68%)

Figure 1 shows that 68% of customers in the sample did not churn, while 32% did. This moderately imbalanced distribution is common in real-world churn datasets.

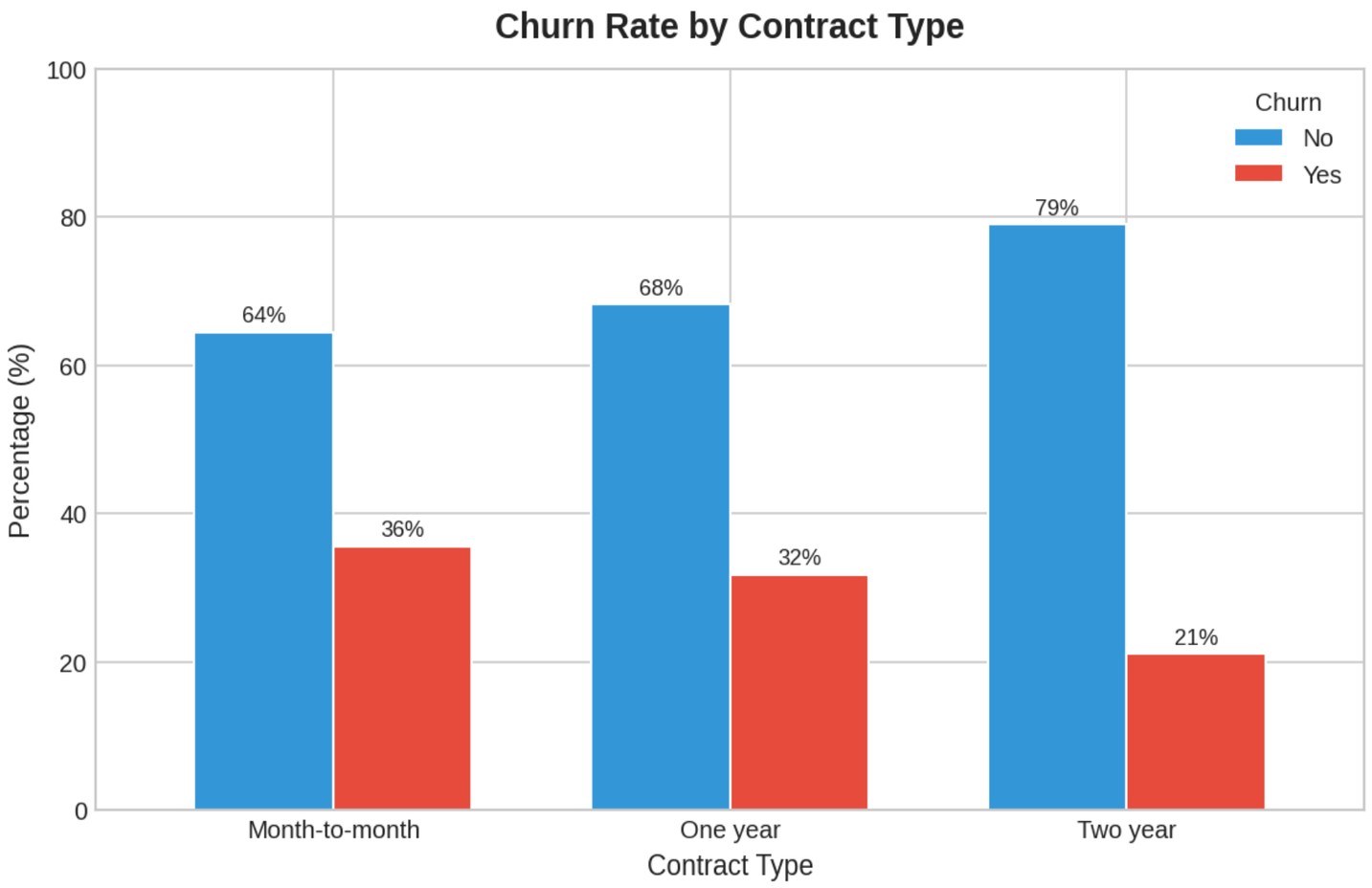


Figure 2: Churn Rate by Contract Type

Figure 2 clearly indicates that customers on Month-to-month contracts have a substantially higher churn rate compared to those on One-year or Two-year contracts. This finding is consistent with industry observations.

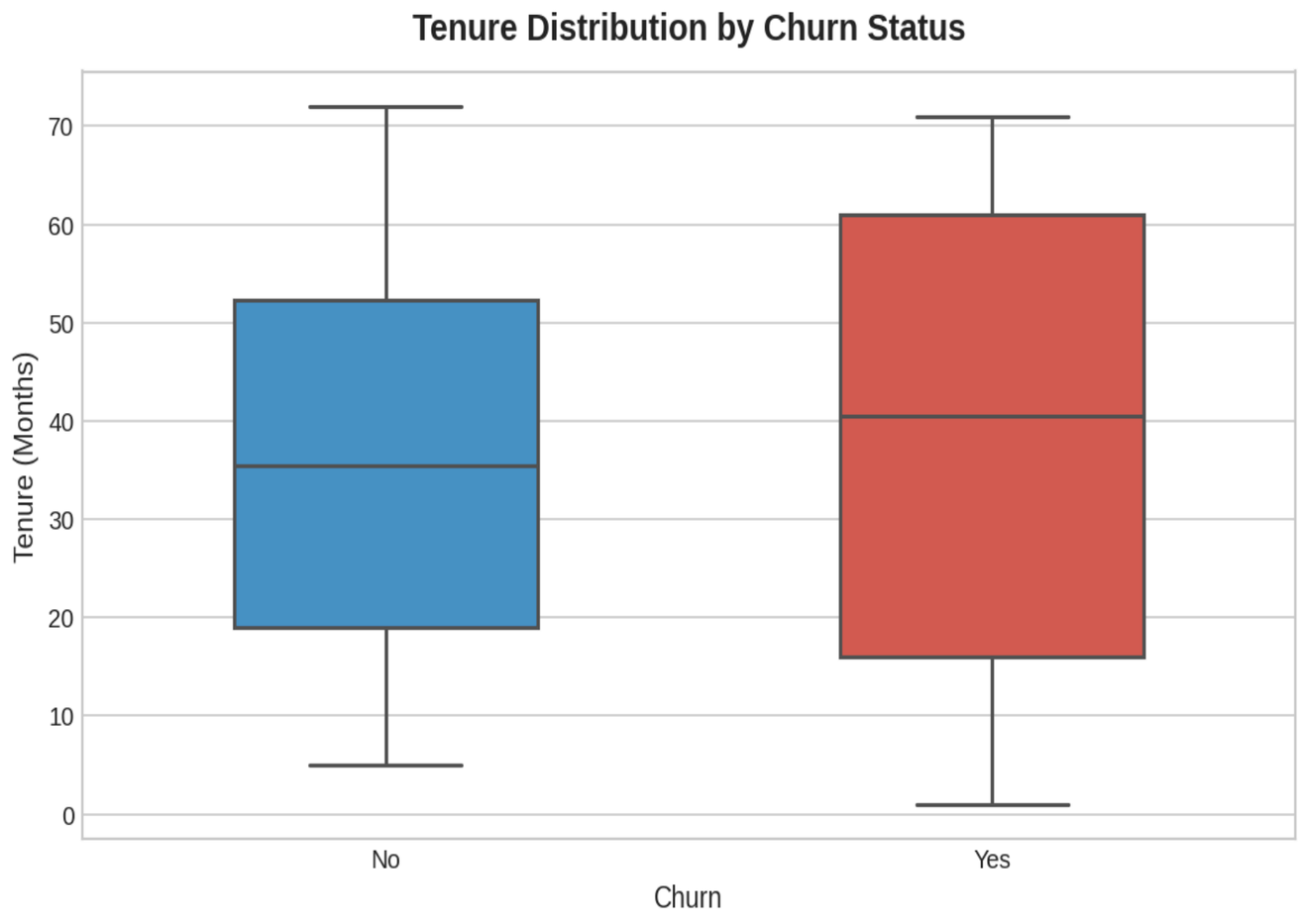


Figure 3: Tenure Distribution by Churn Status

Figure 3 shows that customers who churned (Yes) generally have lower median tenure compared to those who stayed (No). Longer tenure is associated with lower churn risk.

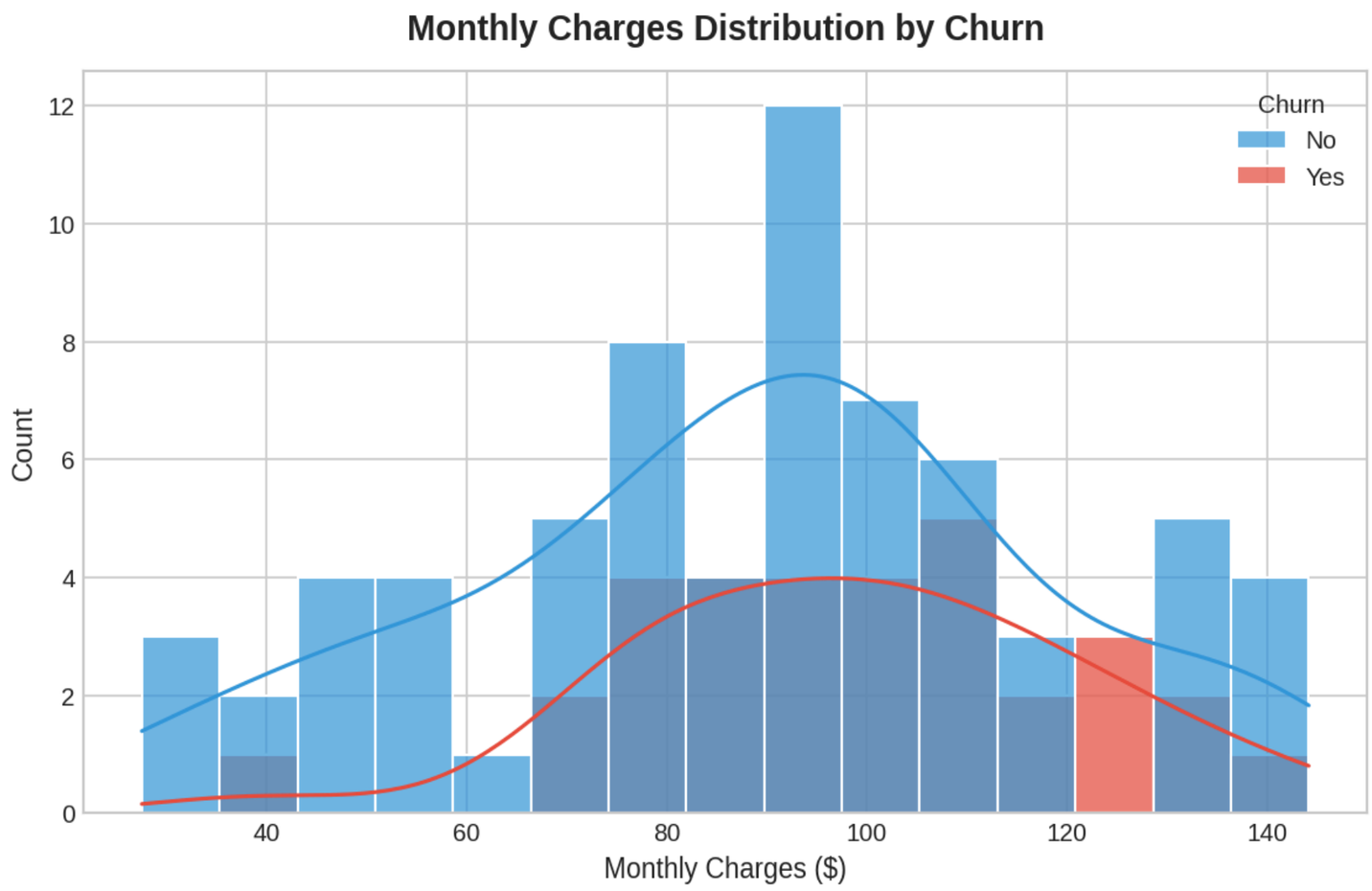


Figure 4: Monthly Charges Distribution by Churn Status

Figure 4 illustrates the distribution of monthly charges for both churned and retained customers. Higher monthly charges appear more frequently among customers who churned.

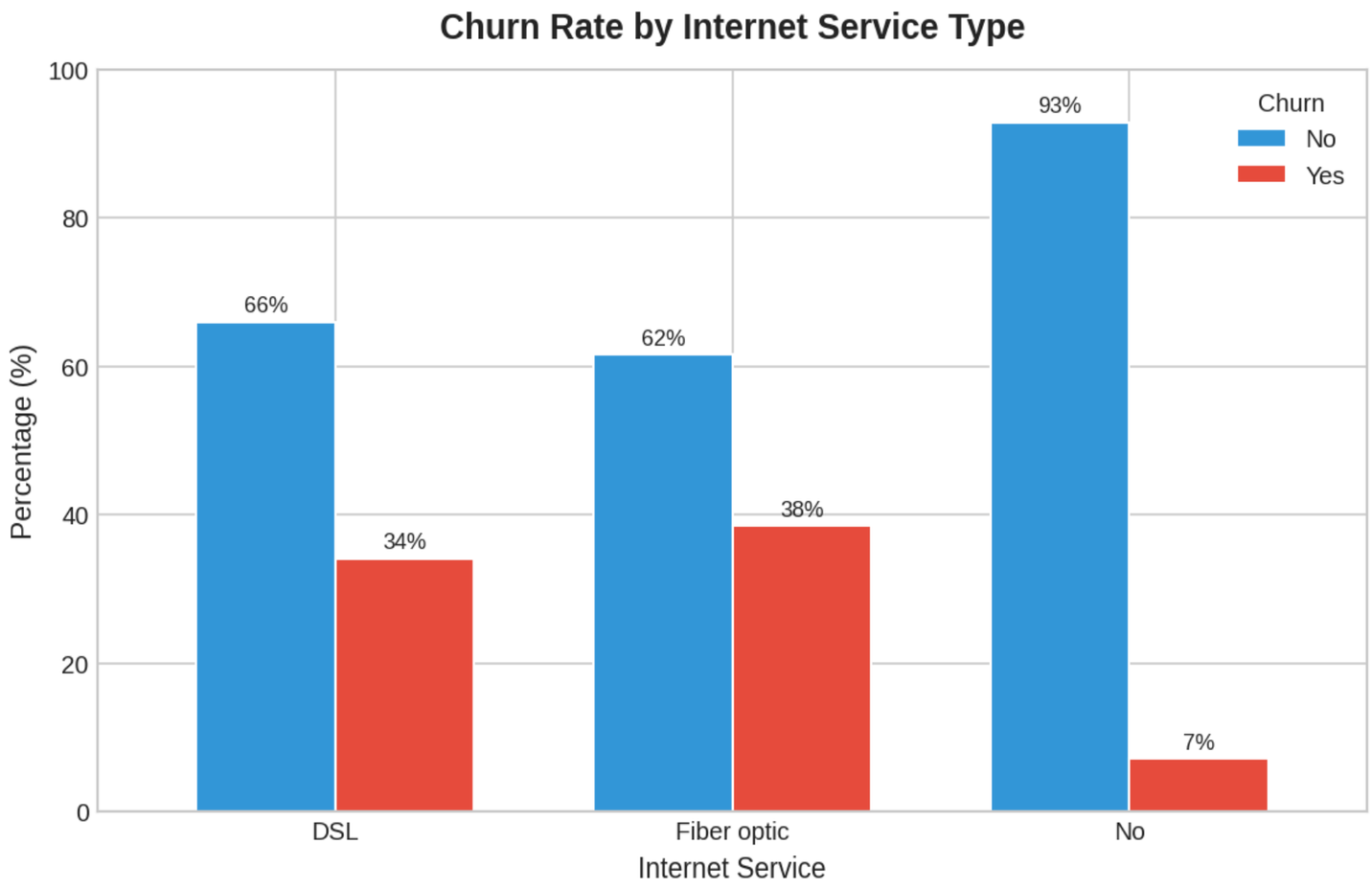


Figure 5: Churn Rate by Internet Service Type

Figure 5 shows variation in churn rates across internet service types. Fiber optic customers tend to show relatively higher churn in this sample, which may relate to pricing or service expectations.

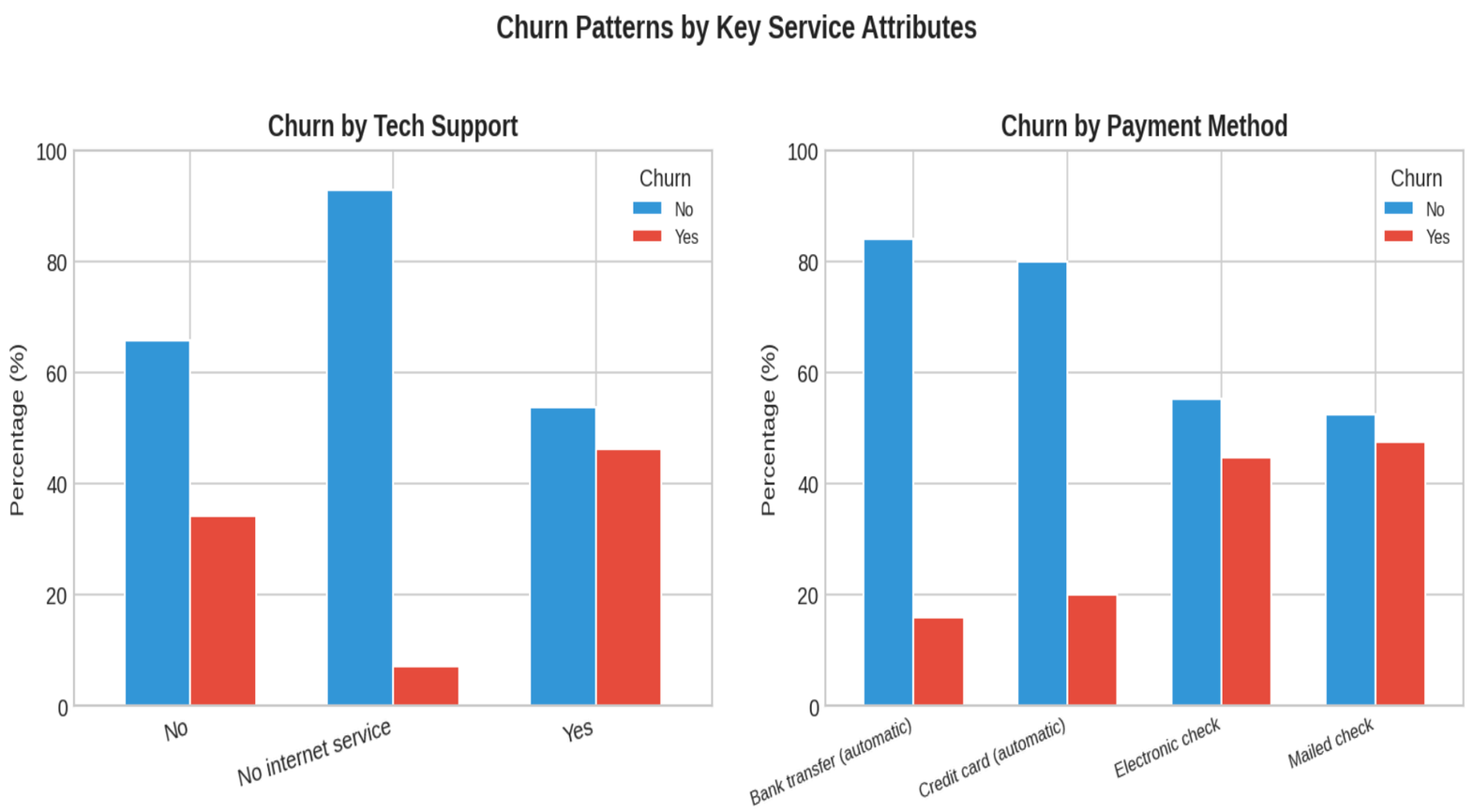


Figure 6: Churn Patterns by Tech Support and Payment Method

Figure 6 highlights additional patterns: customers without technical support and those using electronic check as a payment method tend to have higher churn rates in the sample data.

3.6 Data Analysis Approach

The dataset will be imported into IBM SPSS Modeler for data inspection, preparation, and predictive model development. Relevant variables will be examined to identify patterns associated with customer churn. A suitable classification algorithm will then be selected and evaluated using appropriate performance metrics.

The results will be interpreted carefully because the dataset is synthetic and is intended only to demonstrate the predictive analytics process.

4. METHODS, TOOLS & TECHNOLOGIES USED

4.1 Predictive Analytics

Predictive analytics is a data analysis technique used to identify patterns in existing data and predict possible outcomes. In this project, predictive analytics is used to analyse customer-related information and predict customer churn. The objective is to understand patterns associated with customers leaving an organization and demonstrate how data-driven predictions can support Customer Relationship Management.

4.2 IBM SPSS Modeler

IBM SPSS Modeler is a data mining and machine learning software used for data preparation, data exploration, predictive modelling, and evaluation. In this project, IBM SPSS Modeler is used to import the customer dataset, examine customer attributes, build a classification model, and generate churn predictions.

4.3 Dataset and Variables

The project uses a synthetic customer dataset containing 100 customer records and 16 attributes. The dataset has been created for academic and demonstration purposes and does not represent actual customers of a company.

The important variables used in the project include Tenure, MonthlyCharges, Contract, InternetService, TechSupport, PaymentMethod, PaperlessBilling, and Churn (target).

4.4 Data Preparation

Data preparation involves checking the dataset structure, verifying field types, identifying missing or inconsistent values, and assigning appropriate roles to variables. CustomerID is treated as an identifier, while Churn is selected as the target variable.

4.5 CHAID Decision Tree Algorithm

The CHAID (Chi-squared Automatic Interaction Detection) Decision Tree is a classification algorithm that creates a tree-like model using decision rules derived from chi-square tests. In this project, CHAID is used to classify customer records into the Churn categories Yes and No.

4.6 Model Evaluation Techniques

Model evaluation measures that may be used include Accuracy, Precision, Recall, and Confusion Matrix, depending on the available software outputs.

5. WORKFLOW

The workflow of the Customer Churn Prediction project consists of a series of steps, starting from dataset import and ending with prediction and model evaluation.

5.1 Dataset Import

The customer dataset containing 100 records and 16 attributes is prepared in Excel and imported into IBM SPSS Modeler for analysis.

5.2 Data Preparation

The dataset is checked to verify field names, data types, and data consistency. CustomerID is treated as an identifier, and Churn is selected as the target variable.

5.3 Data Exploration

The dataset is explored using visualizations (see Figures 1– 6) and statistical summaries to understand customer characteristics and churn patterns.

5.4 Predictive Model Building

A CHAID Decision Tree classification model is selected in IBM SPSS Modeler. The model uses relevant customer attributes to classify records into the Churn categories Yes and No.

5.5 Prediction Generation

After the model is executed, predictions are generated for customer records.

5.6 Model Evaluation

The model’ s performance is evaluated using available classification results such as a confusion matrix, accuracy, precision, and recall.

5.7 Workflow Diagram

Customer Dataset → Dataset Import → Data Preparation → Data Exploration → Decision Tree Model Building → Prediction Generation → Model Evaluation → Findings and Recommendations

6. OBJECTIVES OF THE STUDY

The main objectives of this study are:

1. To understand customer churn: To study the concept of customer churn and understand the importance of predicting customer turnover in Customer Relationship Management.

2. To develop a predictive model: To use IBM SPSS Modeler and the CHAID Decision Tree algorithm to analyze customer data and explore how predictive analytics can

support customer churn prediction.

3. To identify key factors: To examine which customer attributes (such as contract type, tenure, and monthly charges) are most strongly associated with the likelihood of churn.

4. To evaluate model performance: To assess the classification performance of the developed model using appropriate evaluation metrics.

7. SCOPE AND SIGNIFICANCE OF THE STUDY

1. Scope of the Study

The scope of this study is to explore the application of predictive analytics in customer churn prediction using IBM SPSS Modeler. The study uses a synthetic dataset containing 100 customer records and 16 attributes.

The study focuses on understanding customer churn, exploring data, applying the CHAID algorithm, and interpreting model outputs for learning purposes.

2. Significance of the Study

Customer churn can create challenges including loss of recurring revenue and higher acquisition costs. Predictive analytics can support CRM decision-making, early identification of churn patterns, practical learning of IBM SPSS Modeler, and improved retention planning.

8. MODEL OUTPUT AND RESULTS

1. Introduction

This section presents the outcome of applying the CHAID Decision Tree algorithm in IBM SPSS Modeler on the synthetic customer dataset of 100 records and 16 attributes. The target variable is Churn (Yes/No).

2. Dataset Summary

The distribution of the target variable is shown below and visualized in Figure 1.

Churn Category	Number of Customers

No	68
Yes	32
Total	**100**

The dataset contains 68 records in the “No” category and 32 records in the “Yes” category (32% churn rate).

3. CHAID Decision Tree Model Output

The CHAID Decision Tree algorithm was applied using IBM SPSS Modeler. The model output displayed a decision tree with a depth of 2 and a total of 5 nodes. Primary splits commonly occurred on variables such as Contract type and Tenure, which aligns with the patterns observed in the exploratory charts (Figures 2 and 3).

4. Interpretation of Results

Key observations from the analysis and visualizations:

- The dataset contains 100 customer records with a 32% churn rate.
- Month-to-month contracts show the highest association with churn (Figure 2).
- Customers who churn tend to have shorter tenure (Figure 3).
- Higher monthly charges are more common among churned customers (Figure 4).
- Absence of Tech Support and use of Electronic check payment method are associated with higher churn (Figure 6).
- The CHAID tree had a depth of 2 and 5 nodes, providing a simple but interpretable model structure.

5. Limitations in Interpreting the Model Results

Results should be interpreted carefully because the analysis uses a synthetic dataset. Identification fields such as CustomerID must not be used as predictors. Further refinement and proper validation (accuracy, precision, recall, confusion matrix) are required before any real-world application.

6. Conclusion of Model Output

The project demonstrates the process of developing and examining a CHAID Decision

Tree model in IBM SPSS Modeler, supported by clear exploratory visualizations. The results are preliminary and educational rather than evidence of a validated production system.

9. RECOMMENDATIONS AND SUGGESTIONS

Based on the study, the following recommendations are proposed:

1. Use of Predictive Analytics in CRM: Organizations should explore predictive analytics to identify patterns associated with customer churn.

2. Focus on Customer Retention: Develop loyalty programs, personalized offers, and proactive support, especially for month-to-month customers.

3. Selection of Relevant Variables: Exclude identification fields such as CustomerID from predictors.

4. Improve Data Quality: Maintain accurate and complete customer data.

5. Validate Model Performance: Use accuracy, precision, recall, and confusion matrix.

6. Target High-Risk Segments: Prioritize customers on month-to-month contracts, short tenure, or lacking tech support.

7. Maintain Customer Data Privacy: Handle customer information responsibly and ethically.

8. Further Improvement: Future studies should use larger, authentic datasets and additional variables.

Overall Recommendation

Predictive analytics can be a useful decision-support tool for CRM. However, model predictions should be combined with professional judgment and proper validation.

10. LIMITATIONS OF THE STUDY

The present study has the following limitations:

1. Limited Dataset Size: Only 100 customer records are used.

2. Use of Synthetic Data: Findings may not reflect actual customer behaviour.

3. Limited Predictive Variables: Other factors such as satisfaction scores or competitor

offers are not included.

4. Model Performance Validation: Full validation with independent test data was not completed.

5. Limited Model Complexity: The decision tree structure is relatively simple.

6. Variable Selection: Care is required to exclude non-predictive attributes.

7. Limited Generalizability: Results cannot be directly generalized to all organizations.

Despite these limitations, the study provides a basic understanding of predictive analytics applied to customer churn using IBM SPSS Modeler.

11. CONCLUSION

The study titled "Customer Churn Prediction Using IBM SPSS Modeler" focuses on understanding the application of predictive analytics in Customer Relationship Management. Customer churn is an important concern because it affects recurring revenue, customer lifetime value, and acquisition costs.

A synthetic dataset of 100 customer records and 16 attributes was analyzed using the CHAID Decision Tree algorithm. Exploratory visualizations (Figures 1– 6) revealed clear patterns: higher churn among month-to-month contracts, shorter tenure, higher charges, and lack of tech support.

The generated decision tree had a depth of 2 and five nodes. The project provided practical learning in data preparation, visualization, predictive modelling, and interpretation using IBM SPSS Modeler.

Due to the small synthetic dataset and need for further validation, the results should be considered preliminary. Predictive analytics has strong potential to support CRM by helping organizations identify churn patterns and design effective retention strategies.

12. BIBLIOGRAPHY

1. IBM. IBM SPSS Modeler Documentation. IBM Official Documentation.

2. IBM. IBM SPSS Modeler: Data Mining and Predictive Analytics. IBM Official Resources.

3. Han, J., Kamber, M., & Pei, J. Data Mining: Concepts and Techniques. Morgan Kaufmann.

4. James, G., Witten, D., Hastie, T., & Tibshirani, R. An Introduction to Statistical Learning. Springer.

5. Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. Multivariate Data Analysis. Cengage Learning.

6. IBM. Predictive Analytics and Decision Tree Modelling Resources. IBM Official Resources.

7. Verbeke, W., Martens, D., Mues, C., & Baesens, B. (2011). Building comprehensible customer churn prediction models with advanced rule induction techniques. Expert Systems with Applications.